Shantanu Kumar Singh

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Education

Birla Institute of Technology (BIT), Mesra — Ranchi, Jharkhand
B.Tech in Electrical and Electronics Engineering – CGPA: 7.75 — *2024 – 2028*

St. Patrick's Sr. Sec. School, Jaunpur
Class XII (Senior Secondary) – 86.6% — *2023*

Experience

Parikshak.ai *(AI EdTech Startup)* — May 2026 – July 2026
Software Engineering Intern — *Founder's Office*

- Engineered a hybrid delay-routing architecture with product and ML teams in Agile sprints, tiering SQS ($\leq$15-minute delays) with EventBridge Scheduler (up to 4 hours) and validating a 5-attempt retry sequence to bypass the SQS delay ceiling.
- Deployed an ARM64 Lambda microservice as an ECR container image provisioned via AWS CDK, unifying 5 services (Lambda, SQS, EventBridge, S3, ECR) into one reproducible production environment.
- Owned pre-launch black-box QA across every product module, running positive, negative, and edge-case test flows against staging to gate each production release, documenting defects for the development team.
- Hardened reliability and security: eliminated 502 cold-start errors by deferring MongoDB initialization to a non-blocking task traced via CloudWatch; blocked duplicate records under concurrent SQS delivery with a 3-field compound unique index and atomic upserts; and rejected unauthorized webhooks via HMAC-SHA1 validation at ingress.

Projects

Distributed Rate Limiter | *Go, Redis, Lua, nginx, Docker, k6* | Source Code — Jul 2026

- Built a horizontally scaled API rate limiter across 3 stateless Go replicas behind nginx, enforcing per-tenant quotas via an O(1) single-round-trip atomic Lua script in Redis that collapses read-decide-write into one operation, eliminating the cross-replica lost-update race without distributed locking.
- Shipped 4 interchangeable limiter algorithms behind a pluggable Go interface (fixed window, sliding log, sliding window counter, token bucket), selectable per tenant at runtime via config with zero redeploys.
- Validated correctness under a k6 load harness: naive per-replica in-memory counters overshot a 100 req/s cap by 3×, while the atomic Lua token bucket held exactly at the configured limit through lazy refill.

Niyantran | *TypeScript, React, Node.js, FastAPI, PostgreSQL, Redis, BullMQ, Groq, Docker* | Source | Live — Jun 2026

- Architected a full-stack AI Traffic Command Center with role-based access control (controller/fleet), a 10-stage end-to-end planning pipeline, and O(V+E) BFS congestion propagation over a 294-junction Bengaluru road graph — broadcasting live WebSocket heatmap updates every 30 seconds to concurrent controller sessions.
- Engineered a Congestion Propagation Engine on BullMQ/Redis job queues, applying an M/M/c/K queueing model to detect corridor spillback and generate heatmaps at 3 time horizons (T+5/15/30 min), with 1.5 km radius conflict resolution when multiple events collide.
- Integrated an MLflow-tracked CatBoost and LightGBM stacking ensemble with conformal prediction intervals for uncertainty-aware severity forecasting.

Simple Social | *FastAPI, Async SQLAlchemy, JWT, ImageKit, Streamlit* | Source Code — Apr 2026

- Built an async FastAPI social backend sustaining sub-80 ms p95 on feed and upload endpoints, with JWT bearer-token auth and ownership-based access control via fastapi-users over SQLAlchemy (aiosqlite).

Technical Skills

Languages: C, C++, Python, Go, JavaScript, TypeScript, SQL, HTML, CSS
Backend & Full-Stack: Node.js, Express, FastAPI, React, REST APIs, Async/Await, Webhooks, JWT Auth
Databases & Storage: SQL, MySQL, MongoDB, PostgreSQL, Redis, SQLAlchemy (async)
Cloud & Infrastructure: AWS (Lambda, SQS, EventBridge, S3, ECR, EC2, CloudWatch), Docker, CDK, CI/CD
Testing & Quality: Black-Box & Functional QA, Regression Testing, Staging Validation, Load Testing (k6), Postman
Developer Tools: Git, GitHub, Linux/Unix, VS Code, Claude Code
CS Fundamentals: Data Structures & Algorithms, Object-Oriented Design & Programming, Complexity Analysis, Concurrency, DBMS, Computer Networks

Achievements

Flipkart: Qualified for Round 3, GRID 8.0; qualified for Round 2, Gridlock 2.0 (AI traffic management, 4,500+ teams).
Competitive Programming: 850+ problems solved; maximum rating 1790 on LeetCode and 1268 (Pupil) on Codeforces.
Contest Ranks: Global Rank 1994 in LeetCode Weekly Contest 489; Global Rank 2830 in Codeforces Round 1090 (Div. 4).
Leadership: Design Head, IET Coding Club, BIT Mesra; Best Speaker, Model United Nations (inter-college).
Athletics: District-level Gold Medallist in Kho Kho and Kabaddi, representing school at inter-school championships.